A Superfast Direct Solver for Nonuniform Discrete Fourier Transform of Type-III

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NUDFT

Nonuniform discrete Fourier transform (NUDFT):

$$f_j = \sum_{k=0}^{N-1} c_k \mathrm{e}^{-2\pi \mathrm{i} \omega_k x_j}, \quad 0 \le j \le M-1.$$

- Sample points: $\{x_j\}_{j=0}^{M-1} \subset [0, 1)$.
- Frequencies: $\{\omega_k\}_{k=0}^{N-1} \subset [-1/2, N-1/2)$.
- Coefficients: $\{c_k\}_{k=0}^{N-1} \subset \mathbb{C}$.
- Target values: $\{f_j\}_{j=0}^{M-1} \subset \mathbb{C}$.
- NUDFT matrix: $\mathbf{A} = (\mathrm{e}^{-2\pi \mathrm{i} x_j \omega_k})_{j,k} \in \mathbb{C}^{M \times N}$.

Forward and Inverse Problems

Forward problem: $\mathbf{f} = \mathbf{A}\mathbf{c}$.

- Uniform case (FFT): $M = N$, $x_j = j/M$ and $\omega_k = k$.
  Complexity: $\mathscr{O}(N \log N)$ [Cooley and Tukey 1965].
- Nonuniform case (NUFFT)
  - Type-I: Equispaced sample points, i.e., $x_j = j/M$;
  - Type-II: Continuous integer frequencies, i.e., $\omega_k = k$;
  - **Type-III**: Both sample points and frequencies are nonuniform.

  Complexity: $\mathscr{O}(M + N + N \log N)$ [Barnett, Magland, and Klinteberg 2019].

Inverse problem: $\mathbf{c} = \mathbf{A}^{\dagger}\mathbf{f}$.

- Iterative methods: CG + NUFFT.
- Direct methods: [Kircheis and Potts 2019], [Wilber, Epperly, and Barnett 2025].

Displacement Structure of NUDFT-II Matrix

Method in [Wilber, Epperly, and Barnett 2025]:
Type-II NUDFT matrix: $\mathbf{A} = (\mathrm{e}^{-\mathrm{i}2\pi k x_j})_{j,k} \in \mathbb{C}^{M\times N}$.
DFT matrix: $\mathbf{F} = (\mathrm{e}^{-2\pi \mathrm{i} k\ell/N})_{k,\ell} \in \mathbb{C}^{N\times N}$.
Let $\tilde{\mathbf{A}} = \mathbf{A}\mathbf{F}^{-1}$ then $\tilde{\mathbf{A}}$ satisfies the following Sylvester equation:

$$\boldsymbol{\Gamma}\tilde{\mathbf{A}} - \tilde{\mathbf{A}}\mathbf{D} = \mathbf{u}\mathbf{v}^\top.$$

- $\boldsymbol{\Gamma} = \mathrm{diag}(\gamma_0, \gamma_1, \ldots, \gamma_{M-1})$ with $\gamma_j = \mathrm{e}^{-2\pi \mathrm{i} x_j}$;
- $\mathbf{D} = \mathrm{diag}(\xi_0, \xi_1, \ldots, \xi_{N-1})$ with $\xi_k = \mathrm{e}^{-2\pi \mathrm{i} k/N}$;
- $u_j = \gamma_j^N - 1$;
- $v_k = \xi_k / N$.

$\rightsquigarrow$ $\tilde{\mathbf{A}}$ has a displacement structure with displacement rank 1.

Displacement Structure of NUDFT-II Matrix

Sylvester equation:

$$\mathbf{\Gamma}\tilde{\mathbf{A}} - \tilde{\mathbf{A}}\mathbf{D} = \mathbf{u}\mathbf{v}^\top.$$

For row and column index sets $\mathcal{J}$ and $\mathcal{K}$, the submatrix $\tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}}$ satisfies:

$$\mathbf{\Gamma}_{\mathcal{J}}\tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}} - \tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}}\mathbf{D}_{\mathcal{K}} = \mathbf{u}_{\mathcal{J}}\mathbf{v}_{\mathcal{K}}^\top.$$

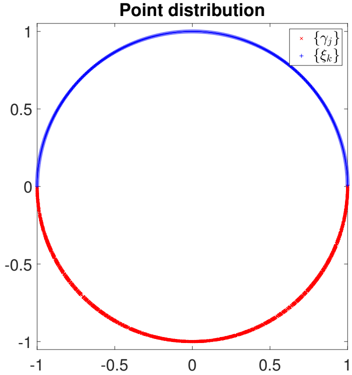
When $\{\gamma_j\}_{j\in\mathcal{J}}$ and $\{\xi_k\}_{k\in\mathcal{K}}$ are disjoint:

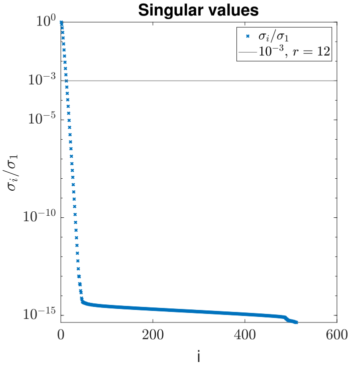
- $\rightsquigarrow$ The singular values of $\tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}}$ decay exponentially [Beckermann and Townsend 2017];
- $\rightsquigarrow$ **Low-rank property:** $\tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}}$ can be approximated by a low-rank matrix.

Low-Rank Property in the NUDFT-II Matrix

$\tilde{\mathbf{A}}$ is a Cauchy-like matrix:

$$\tilde{\mathbf{A}}(j,k) = \frac{u_j v_k}{\gamma_j - \xi_k}.$$





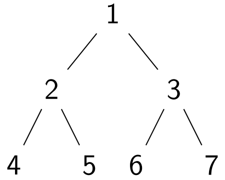
HSS Matrix

Idea: Approximate $\tilde{\mathbf{A}}$ by a hierarchical semiseparable (HSS) matrix.

Level 0

Level 1

Level 2



$\mathcal{K}_1 = [0, \ldots, 255]$

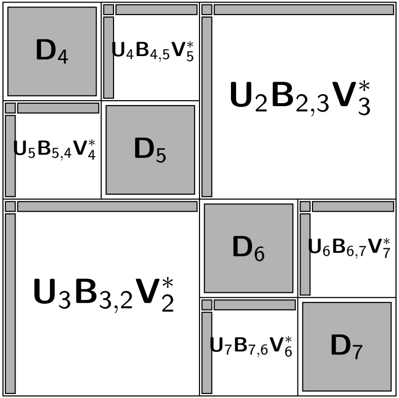
$\mathcal{K}_2 = [0, \ldots, 127]$, $\mathcal{K}_3 = [128, \ldots, 255]$

$\mathcal{K}_4 = [0, \ldots, 63]$, $\mathcal{K}_5 = [64, \ldots, 127]$, …

$$\mathbf{D}_1 = \begin{bmatrix} \mathbf{D}_2 & \mathbf{U}_2\mathbf{B}_{2,3}\mathbf{V}_3^{*} \\ \mathbf{U}_3\mathbf{B}_{3,2}\mathbf{V}_2^{*} & \mathbf{D}_3 \end{bmatrix},$$

$$\text{recursion on the diagonal:} \quad \mathbf{D}_2 = \begin{bmatrix} \mathbf{D}_4 & \mathbf{U}_4\mathbf{B}_{4,5}\mathbf{V}_5^{*} \\ \mathbf{U}_5\mathbf{B}_{5,4}\mathbf{V}_4^{*} & \mathbf{D}_6 \end{bmatrix},$$

$$\text{recursion on the off-diagonal:} \quad \mathbf{U}_2 = \begin{bmatrix} \mathbf{U}_4\mathbf{R}_4 \\ \mathbf{U}_5\mathbf{R}_5 \end{bmatrix} \quad \text{and} \quad \mathbf{V}_2 = \begin{bmatrix} \mathbf{V}_4\mathbf{W}_4 \\ \mathbf{V}_5\mathbf{W}_5 \end{bmatrix}.$$



$$\mathbf{U}_2 = \begin{bmatrix} \mathbf{U}_4\mathbf{R}_4 \\ \mathbf{U}_5\mathbf{R}_5 \end{bmatrix}, \quad \mathbf{V}_2 = \begin{bmatrix} \mathbf{V}_4\mathbf{W}_4 \\ \mathbf{V}_5\mathbf{W}_5 \end{bmatrix}.$$

$$\mathbf{U}_3 = \begin{bmatrix} \mathbf{U}_6\mathbf{R}_7 \\ \mathbf{U}_6\mathbf{R}_7 \end{bmatrix}, \quad \mathbf{V}_3 = \begin{bmatrix} \mathbf{V}_6\mathbf{W}_6 \\ \mathbf{V}_7\mathbf{W}_7 \end{bmatrix}.$$

Shared basis property: $\mathbf{U}_\tau$ is the column basis of $\tilde{\mathbf{A}}_{\mathrm{HSS}}(\mathcal{J}_\tau, \mathcal{K}_\tau^{\mathrm{c}})$:

$$\tilde{\mathbf{A}}_{\mathrm{HSS}}(\mathcal{J}_4, :) = \begin{bmatrix} \mathbf{D}_4 & \mathbf{U}_4\mathbf{B}_{4,5}\mathbf{V}_5^* & \mathbf{U}_4\mathbf{R}_4\mathbf{B}_{2,3}\mathbf{W}_6^*\mathbf{V}_6^* & \mathbf{U}_4\mathbf{R}_4\mathbf{B}_{2,3}\mathbf{W}_7^*\mathbf{V}_7^*. \end{bmatrix}$$

A Superfast Direct Solver for Type-II NUDFT Problem

Algorithm A direct solver for type-II INUDFT problem.

1: Approximate $\tilde{\mathbf{A}} = \mathbf{A}\mathbf{F}^{-1}$ by an HSS matrix $\tilde{\mathbf{A}}_{\mathrm{HSS}}$.
2: Solve $\mathbf{v} = \arg\min_{\mathbf{w}} \|\tilde{\mathbf{A}}_{\mathrm{HSS}}\mathbf{w} - \mathbf{f}\|_2$ by the HSS least-squares solver.
3: Compute $\mathbf{c} = \mathbf{F}^{-1}\mathbf{v}$ using the iFFT.

Type-II approximation: $\mathbf{A} = \tilde{\mathbf{A}}\mathbf{F} \approx \tilde{\mathbf{A}}_{\mathrm{HSS}}\mathbf{F} =: \mathbf{A}_{\mathrm{fast}}$.
HSS rank: $r = \mathscr{O}(\log(1/\varepsilon)\log(N))$ [Wilber, Epperly, and Barnett 2025].
Construction: Factorized alternating direction implicit (fADI) [Benner, Li, and Truhar 2009].
Complexity: $\mathscr{O}\big((M+N)r^2\big)$.
HSS least-squares solver: URV factorization [Xi et al. 2014].
Complexity: $\mathscr{O}\big((M+N)r^2\big)$ for the factorization and $\mathscr{O}\big((M+N)r\big)$ for the each solution.

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Type-II approximation: $\mathbf{A} = \tilde{\mathbf{A}}\mathbf{F} \approx \tilde{\mathbf{A}}_{\mathrm{HSS}}\mathbf{F} =: \mathbf{A}_{\mathrm{fast}}$.

Complexities:

- **Building the solver:** $\mathscr{O}\big((M+N)r^2\big) = \mathscr{O}\big((M+N)\log^2(N)\big)$.
- **Each solution:** $\mathscr{O}\big((M+N)r + N\log(N)\big) = \mathscr{O}\big((M+N)\log(N)\big)$.

Motivation

Idea: Basis expansion.

$$\mathbf{A}(j,k) = \mathrm{e}^{-2\pi \mathrm{i} x_j \omega_k} = \underbrace{\sum_{\ell \in \mathcal{I}_{N;\sigma}} \mathrm{e}^{-2\pi \mathrm{i} x_j \ell} Q_{\ell,k}}_{\textit{Approximation}} + \underbrace{\sum_{\ell \notin \mathcal{I}_{N;\sigma}} \mathrm{e}^{-2\pi \mathrm{i} x_j \ell} Q_{\ell,k}}_{\textit{Error}} .$$

- $\mathcal{I}_{N;\sigma} = \{\ell \in \mathbb{Z} : -(\sigma N - N)/2 \le \ell \le (\sigma N + N)/2 - 1\}$: σ-expanded set;
  Original index set: $\mathcal{I}_N = \{0,1,2,3,4,5,6,7\}$.
  2-expanded index set: $\mathcal{I}_{N;\sigma} = \{-4,-3,-2,-1,0,1,2,3,4,5,6,7,8,9,10,11\}$.
- $Q_{\ell,k} = G(\ell - \omega_k)$ where $G(\omega) = \mathrm{e}^{\pi \mathrm{i}\omega} \frac{\sin(\pi\omega)}{\pi\omega}$: kernel matrix.

Matrix form:

$$\mathbf{A} = \mathbf{B}\mathbf{Q} + \mathbf{E}.$$

Motivation

Matrix form:

$$\mathbf{A} = \mathbf{B}\mathbf{Q} + \mathbf{E}.$$

Proposition (Li and L., 2025)

If $\{x_j\}$ are i.i.d. uniform random variables in $[0,1)$ and $\{\omega_k = k + \delta_k\}$ are perturbations of the continuous integers with $|\delta_k| \le \alpha$ for some constant $\alpha \in [0, 1/2)$. Then $\mathbb{E}\mathbf{E} = \mathbf{0}$ and $\mathbb{E}\|\mathbf{E}\|_{\mathrm{F}}^2 \le \frac{2\sin^2(\pi\alpha)}{\pi^2((\sigma-1)N/2+1/2-\alpha)}\|\mathbf{A}\|_{\mathrm{F}}^2$.

Practical Algorithm

Denoting $\mathbf{H} = \mathbf{B}^{\dagger}\mathbf{A}$, we have

$$\mathbf{H} = \mathbf{Q} + \mathbf{B}^{\dagger}\mathbf{E},$$
$$\mathbf{A} = \mathbf{B}\mathbf{H} + \left(\mathbf{I} - \mathbf{B}\mathbf{B}^{\dagger}\right)\mathbf{E}.$$

- $\mathbf{B}$ is a NUDFT-II matrix;
- $\mathbf{Q}$ is a kernel matrix, can be compressed into an HSS matrix;
- $\mathbf{H}$ can be empirically approximated by an HSS matrix;
- $\mathbf{I} - \mathbf{B}\mathbf{B}^{\dagger}$ is a projection matrix.

A Superfast Direct Solver for Type-III INUDFT Problem

Algorithm A direct solver for type-III INUDFT problem.

1: Construct a type-II direct solver $\mathbf{B}_{\text{fast}} \approx \mathbf{B}$.
2: Compress $\mathbf{B}_{\text{fast}}^{\dagger}\mathbf{A}$ into an HSS matrix $\mathbf{H}_{\text{HSS}}$.
3: Compute $\mathbf{c} = \mathbf{H}_{\text{HSS}}^{-1}\mathbf{B}_{\text{fast}}^{\dagger}\mathbf{f}$.

Type-III approximation: $\mathbf{A} \approx \mathbf{B}_{\text{fast}}\mathbf{H}_{\text{HSS}}$.
Goal: Find an efficient way to construct an HSS compression of $\mathbf{B}_{\text{fast}}^{\dagger}\mathbf{A}$.

- Fast computation on $\mathbf{A}$: NUFFT.
- Fast computation on $\mathbf{B}_{\text{fast}}^{\dagger}$: URV solution and iFFT.

Black-box setting: Only fast matrix-vector multiplication is available.
$\rightsquigarrow$ [Levitt and Martinsson 2024]: Using random samplings.
Complexity: $\mathscr{O}\left(Nr^2 + T_{\text{mult}}r\right)$, where T_{mult} is the complexity of apply $\mathbf{B}_{\text{fast}}^{\dagger}\mathbf{A}$ to a vector.
Empirically, $r = \mathscr{O}(\log(N))$.

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Type-III approximation: $\mathbf{A} \approx \mathbf{B}_{\mathrm{fast}}\mathbf{H}_{\mathrm{HSS}}$.

Complexities:

- **Preliminary:** Construct and factorize $\mathbf{B}_{\mathrm{fast}} \rightsquigarrow \mathcal{O}\big((M+N)\log^2(N)\big)$.
- **Construction:** Compress $\mathbf{H}_{\mathrm{HSS}} \rightsquigarrow \mathcal{O}\big((M+N)\log^2(N)\big)$.
- **Factorization:** Factorize $\mathbf{H}_{\mathrm{HSS}} \rightsquigarrow \mathcal{O}\big(N\log^2(N)\big)$.
- **Each solution:** $\mathbf{c} = \mathbf{H}_{\mathrm{HSS}}^{-1}\mathbf{B}_{\mathrm{fast}}^{\dagger}\mathbf{f} \rightsquigarrow \mathcal{O}\big((M+N)\log(N)\big)$.

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Type-III approximation: $\mathbf{A} \approx \mathbf{B}_{\mathrm{fast}}\mathbf{H}_{\mathrm{HSS}}$.

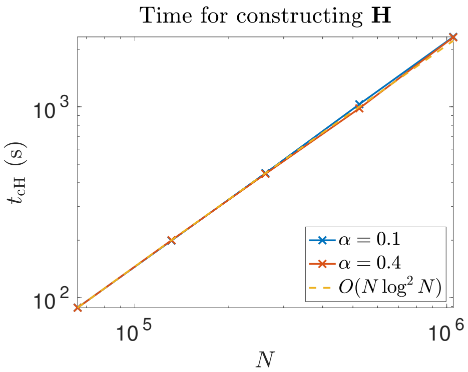
Complexities:

- **Building the solver:** $\mathscr{O}\big((M+N)\log^2(N)\big)$.
- **Each solution:** $\mathscr{O}\big((M+N)\log(N)\big)$.

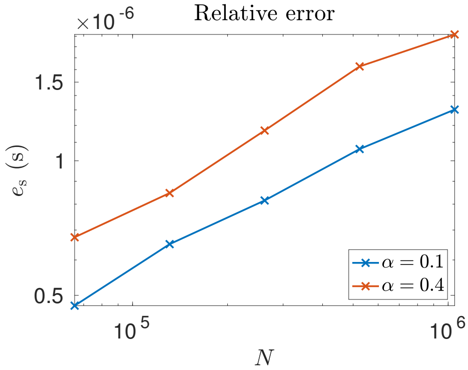
Experiment Setting

- $M = 4N$ and $N = 2^{17}$, 2^{18}, 2^{19}, 2^{20}.
- Samples are i.i.d. uniform random variables in $[0, 1)$.
- Frequencies are perturbated integers given by $\omega_k = k + \alpha\psi_k$ where $\{\psi_k\}$ are i.i.d. uniform random variables on $[-1, 1]$ and $\alpha \in \{0.1, 0.4\}$.

Direct Solver



(a) Construction time on $\mathbf{H}_{\rm HSS}$.



(b) Relative error.

Iterative Solver

N	method	t_{pre}	t_{iter}	n_{iter}	r_{s}	e_{s}
2^{17}	lsqr	-	4.3e+01	500	5.9e-06	1.7e-03
	plsqr	2.3e+02	1.4e+01	5	2.8e-13	1.6e-11
2^{18}	lsqr	-	8.4e+01	500	2.7e-07	1.1e-05
	plsqr	5.2e+02	2.9e+01	5	7.8e-14	4.9e-13
2^{19}	lsqr	-	1.7e+02	500	9.6e-09	3.0e-07
	plsqr	1.2e+03	6.2e+01	5	3.2e-13	1.1e-12
2^{20}	lsqr	-	5.2e+02	500	9.7e-07	6.2e-05
	plsqr	2.6e+03	1.7e+02	5	1.2e-13	7.6e-13

Table: Time and error results of the iterative solver under the random-pertutabtion case, $\alpha = 0.1$.

Iterative Solver

N	method	t_{pre}	t_{iter}	n_{iter}	r_{s}	e_{s}
2^{17}	lsqr	-	4.3e+01	500	5.1e-10	9.1e-09
	plsqr	2.3e+02	2.6e+01	9	8.2e-15	3.2e-14
2^{18}	lsqr	-	8.5e+01	500	1.2e-07	6.3e-06
	plsqr	5.1e+02	5.2e+01	9	9.0e-14	5.6e-13
2^{19}	lsqr	-	2.2e+02	500	2.1e-06	6.4e-04
	plsqr	1.1e+03	9.4e+01	8	2.4e-13	2.6e-11
2^{20}	9 lsqr	-	4.8e+02	500	4.3e-06	5.5e-04
	plsqr	2.6e+03	2.4e+02	9	7.1e-14	1.9e-12

Table: Time and error results of the iterative solver under the random-pertutabtion case, $\alpha = 0.4$.

Conclusions

Conclusions:

- A superfast direct solver for type-III NUDFT problem;
- The direct solver could also be used as an efficient preconditioner;
- Error bound of the forward approximation.

Future work:

- Extension to high-dimensional problems;
- Method for regularized least-squares problems.

My homepage: `https://jingyuliumath.github.io/`.